

Truncation of simplicial sets and applications in simplicial algebras

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Dold-Kan correspondence is not a strong monoidal functor

Let k be a field.

- The Dold-Kan correspondence says there is an equivalence of categories

$$N : s\mathbf{Vect}_k \xrightarrow{\sim} Ch_{\geq 0}(\mathbf{Vect}_k) : \Gamma$$

- Both functors are lax monoidal functors with respect to the tensor product. The equivalence fail to preserve the monoid objects, the simplicial k -algebras and the dg k -algebras.
- So how can we study the simplicial k -algebras in the light of the Dold-Kan correspondence?

Studies in similar directions

- Whitehead and Loday characterized the simplicial groups with normalizing complex of length ≤ 1 as the crossed modules of groups (= strict 2-groups = internal category in $\mathbf{Grp}$).
- Cegarra and Carrasco characterized the simplicial groups/ k -algebras as the hypercrossed complexes.
- Schwede and Shipley showed the Dold-Kan correspondence induces a Quillen equivalence between the simplicial k -algebras and the dg k -algebras.
- Gerstenhaber showed the equivalence classes of crossed modules of groups/ k -algebras correspond to the 3rd group/Hochschild cohomology.
- Conduché generalized the notion of the crossed modules to 2-crossed modules for groups.
- Porter, Arvasi, Casas, Ladra, Wagemann etc. continued the study the crossed modules of k -algebras/Lie algebras.

Main questions

By convention, s.algebra = simplicial unital associative k -algebras and s.vector space = simplicial k -vector spaces.

1. Existence Problem: Can we give a simple criterion to determine on which $\text{s.vector space } V$, there exists some s.algebra ?
2. Classification Problem: Can we classify a certain family of s.algebras ? For instance, the simple s.algebras or the s.algebras on a fixed $\text{s.vector space } V$?

Examples of simplicial objects

Example

Let X be a topological space. The singular chain complex $Sing(X)_n = \mathbb{Z}[\text{Hom}_{\text{Top}}(\Delta^n, X)]$ is a simplicial abelian group.

Example

Let A be an associative k -algebra A and M be a bimodule over A . The Hochschild complex $C(A; M)_n = M \otimes A^{\otimes n}$ is a simplicial k -vector space.

Example

∞ -categories and Kan complexes are simplicial sets.

Simplicial object

Definition

A simplicial object X in a category $\mathcal{C}$ is a collection of objects $\{X_n\}_{n \geq 0}$ in $\mathcal{C}$ with face maps $d_i : X_n \rightarrow X_{n-1}$ and degeneracy maps $s_i : X_n \rightarrow X_{n+1}$ being morphisms in $\mathcal{C}$ for $0 \leq i \leq n$ (except that there are no d_0 on X_0) such that the following simplicial identities hold

$$\begin{aligned} d_i d_j &= d_{j-1} d_i & i < j \\ s_i s_j &= s_{j+1} s_i & i \leq j \\ d_i s_j &= \begin{cases} s_{j-1} d_i & i < j \\ \text{id} & i = j, j+1 \\ s_j d_{i-1} & i > j+1. \end{cases} \end{aligned}$$

The Dold-Kan correspondence

Let $\mathcal{A}$ be an abelian category and X be a simplicial object in $\mathcal{A}$.

$$\cdots \rightrightarrows X_2 \rightrightarrows X_1 \rightrightarrows X_0$$

Definition

The normalizing complex of X is the chain complex

$$NX = (\cdots \rightarrow NX_n \xrightarrow{d_n} \cdots \rightarrow NX_1 \xrightarrow{d_1} X_0)$$

where $NX_n = \bigcap_{0 \leq i < n} \ker d_i$. We define $H_n(X) = H_n(NX)$.

Theorem (Dold, Kan)

$N : s\mathcal{A} \xrightarrow{\sim} Ch_{\geq 0}(\mathcal{A})$ is an equivalence of category with inverse Γ .

The first example of s.algebra

$k\Delta^0$ is the s.vector space that $(k\Delta^0)_n = k$ and all the face and degeneracy maps are the identity maps on k .

Diagrammatically, $k\Delta^0$ is

$$\dots \begin{array}{c} \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \end{array} k \begin{array}{c} \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \end{array} k \begin{array}{c} \leftarrow \\ \leftarrow \\ \leftarrow \end{array} k$$

and the normalizing complex $N(k\Delta^0)$ is $(0 \rightarrow k)$.

The Dold-Kan correspondence says $\Gamma((0 \rightarrow k)) = k\Delta^0$.

In particular, $k\Delta^0$ is also a simple s.algebra.

Another simple s.algebra

Let V be the s.vector space with normalizing complex

$$NV = (0 \rightarrow k \xrightarrow{1} k).$$

By Dold-Kan correspondence, $V_n = \bigoplus_{i=0}^n k\xi_i^n \cong k^{n+1}$.

$$\dots \rightrightarrows k^3 \rightrightarrows k^2 \rightrightarrows k$$

Proposition (HSZZ, 26')

There exists a unique s.algebra B on V where $\xi_i^n \xi_j^n = \xi_j^n \xi_i^n = \xi_i^n$ for all $0 \leq i \leq j \leq n$, with the convention that ξ_n^n is the identity element. Moreover, B is commutative and simple.

Truncation of simplicial sets

Definition (HSZZ, 26')

Let X be a simplicial set and J be a simplicial subset. Let $l \geq 0$. Define the l -th truncation of X based on J to be

$$(T_{l,J})_n = \begin{cases} J_n & n < l \\ \{x \in X_n \mid d_{n \rightarrow l-1}(x) \in J_{l-1}, \forall d_{n \rightarrow l-1}\} & n \geq l \end{cases}$$

with the convention that $T_{0,J} = X$.

Below is what $T_{2,J}$ informally looks like.

$$\cdots \Rrightarrow \Rrightarrow \Rrightarrow \{x \in X_2 : d_i x \in J_1, \forall 0 \leq i \leq 2\} \Rrightarrow \Rrightarrow \Rrightarrow J_1 \Rrightarrow J_0$$

Classification of simple simplicial k -algebras

Lemma (HSZZ 26')

Let A be a s.algebra. Suppose I_0 is a two-sided ideal of A_0 . The truncation T_{l,I_0} is a two-sided ideal of A for all $l \geq 0$.

Theorem (HSZZ 26')

Let A be a simple s.algebra.

1. A_0 is a simple k -algebra.
2. NA concentrates on degree ≤ 1 and is exact.
3. $A \cong S$ or $B \otimes S$ for some simple k -algebra S .
4. If A_0 is of finite dimension, then $A \cong M_t(D)$ for some s.algebra D .

Square-zero extension

Lemma (HSZZ, 26')

Let A be a s.algebra and M be a simplicial bimodule over A . Then there exists a s.algebra structure on $A \oplus M$ given by

$$\begin{aligned}(a, m) + (a', m') &= (a + a', m + m') \\ (a, m) \cdot (a', m') &= (aa', a \cdot m' + m \cdot a').\end{aligned}$$

Example (HSZZ, 26')

$\Gamma(\Sigma^q(0 \rightarrow k \xrightarrow{1} k))$ admits simplicial module structures over B .

k -augmented simplicial algebras

Definition

Let A be a s.algebra. We say A is k -augmented if there exists a morphism of s.algebra $A \rightarrow k\Delta^0$.

Lemma

A s.algebra A is augmented iff there is a k -algebra morphism $\epsilon : A_0 \rightarrow k$ such that ϵ factors through $\text{coker}(d_0 - d_1)$.

Example

$k\Delta^0$ is k -augmented whereas B is not.

Two lemmas

Lemma (HSZZ, 26')

Let A be a k -augmented s.algebra and V be any s.vector space. Then there exists a (commutative) s.algebra structure on $A \oplus V$.

Lemma (HSZZ, 26')

Let A be a s.algebra such that $H_0(A) = 0$. Then any left simplicial (unital) module M over A has exact normalizing complex. In particular, NA is exact.

The existence theroem

Theorem (HSZZ 26')

Let V be a s.vector space with $V_0 \neq 0$. Then there exists a s.algebra structure on V iff NV is exact or $H_0(V) \neq 0$.

Remark

If exists, the s.algebras constructed in our proof are commutative, since they are the square-zero extensions of $k\Delta^0$ and B .

Example

There exists no s.algebra struture on $\Gamma(0 \rightarrow k^3 \twoheadrightarrow k^2)$.

Questions for future studies

1. On which s.vector spaces do noncommutative s.algebras exist?

They may be the path algebras of some simplicial quivers.

We know when noncommutative crossed modules exist.

2. Can we replace the associative k -algebras structure with the structure of coassociative k -coalgebras, Hopf algebras, Frobenius algebras etc.?

We classified the simple s.Lie/ s.Jordan/ etc, algebras

3. How to classify semisimple s.algebras, s.Lie algebras, etc?

Thank you!

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